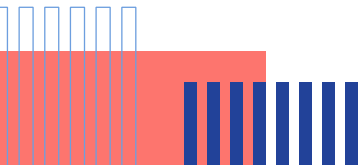




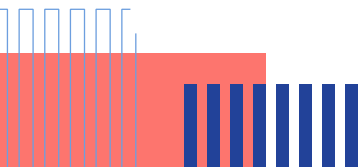
READING BETWEEN THE LINES

Simren Arora, Julica Hatabu, Arjun Selvam, Justin Zeng, Jonathan Zhang





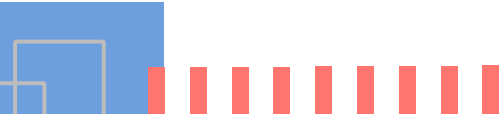
July 3 and 4, 2026



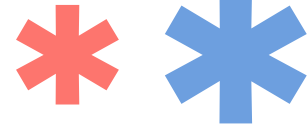


1

Introduction







2

EDA



Dataset

- Kaggle Dataset: News articles for political bias classification

10832
articles

297
sites

2014
to
2025

L LEFT

L LEAN LEFT

C CENTER

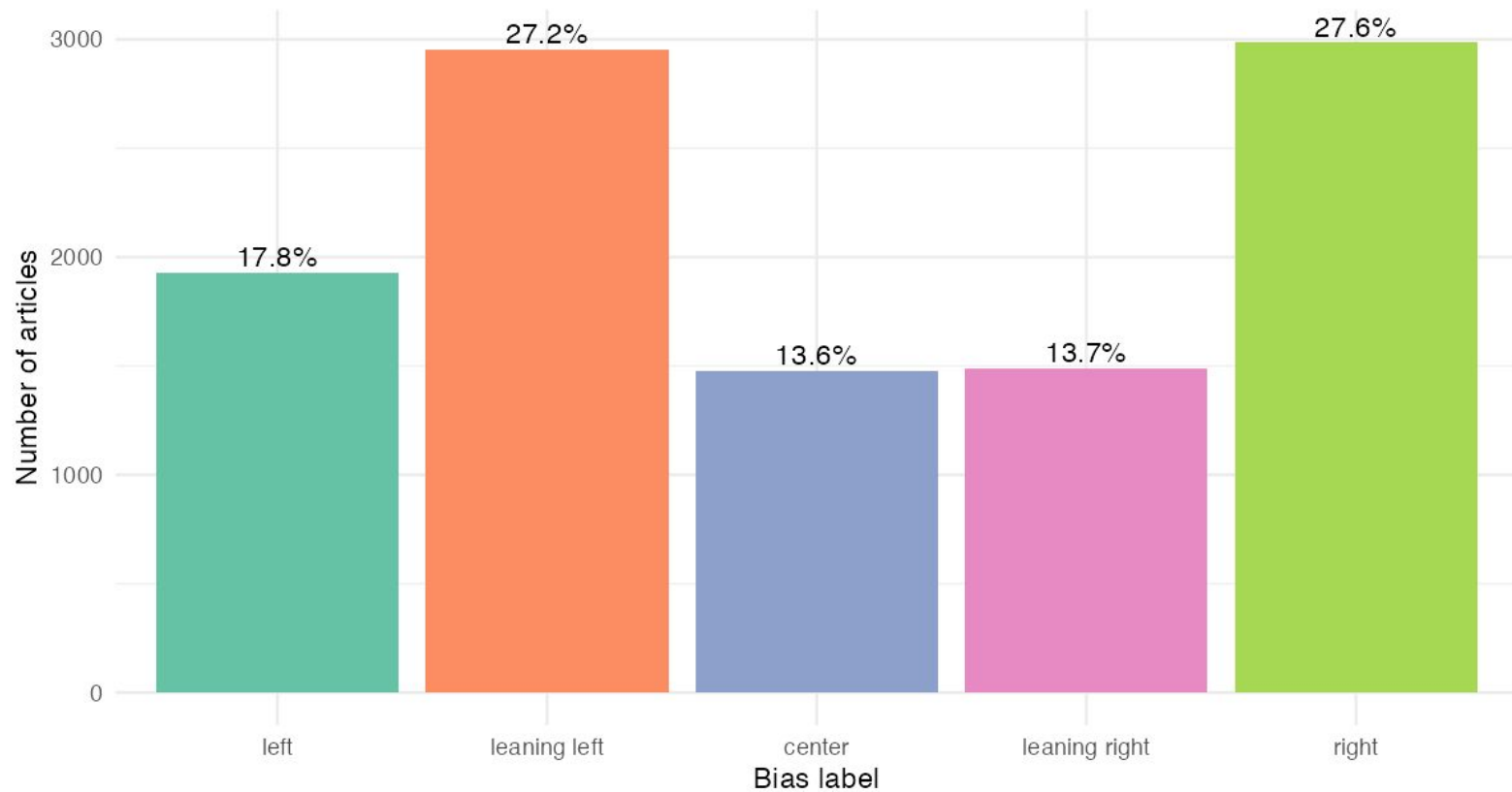
R LEAN RIGHT

R RIGHT

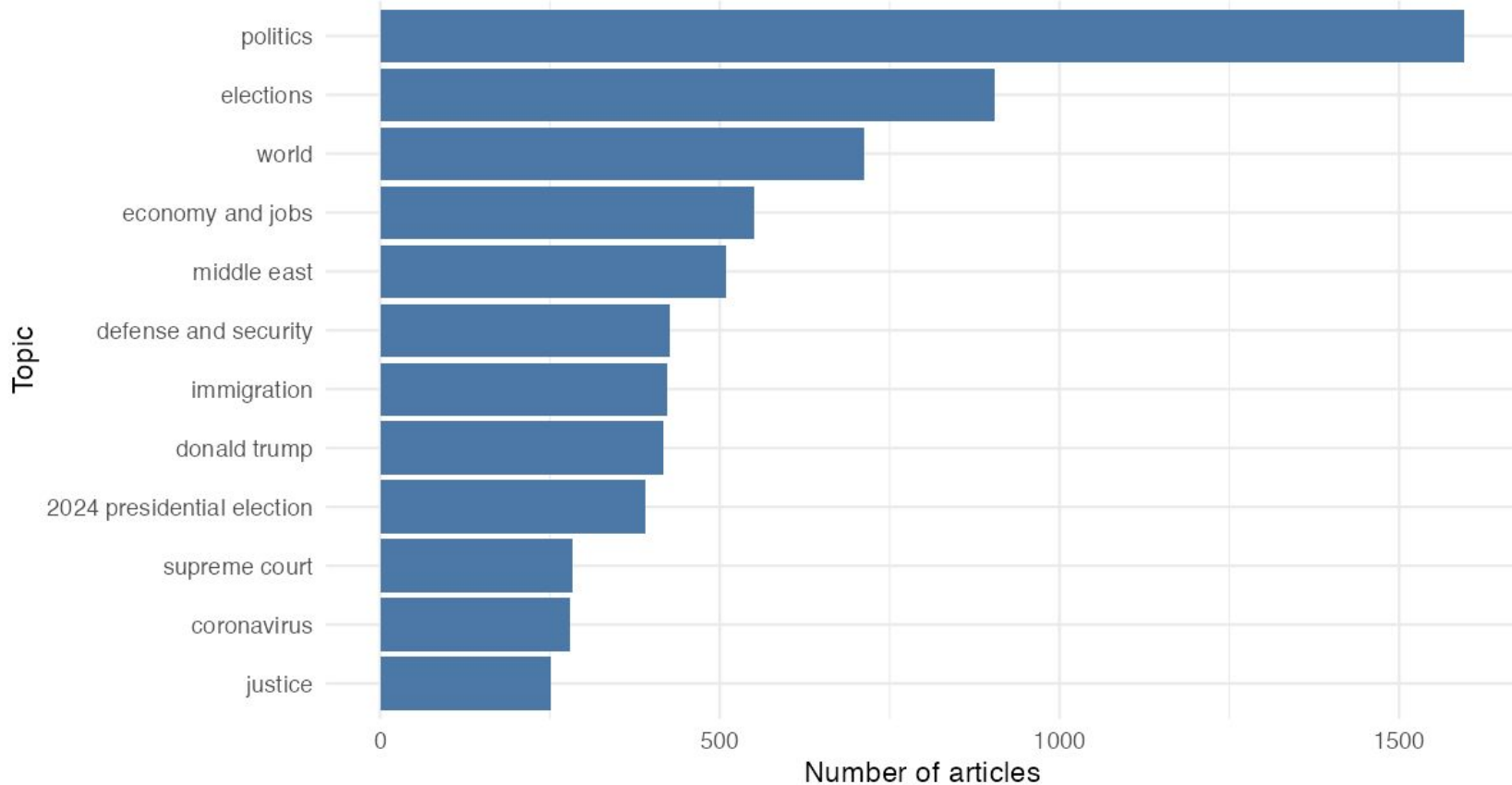
Dataset variables

- **URL:** <https://www.foxnews.com/us/wis-hospital-intentionally-covid-19-vaccine>
- **Date:** 2020-12-31
- **Title:** Wisconsin hospital employee 'intentionally' removed COVID-19 vaccine from refrigeration
- **Site:** Fox News Digital
- **Topic:** coronavirus
- **Bias:** right
- **Page_text:** A Wisconsin-based hospital on Wednesday said an employee who removed 57 vials of the Moderna coronavirus vaccine from the refrigerator earlier this week.....

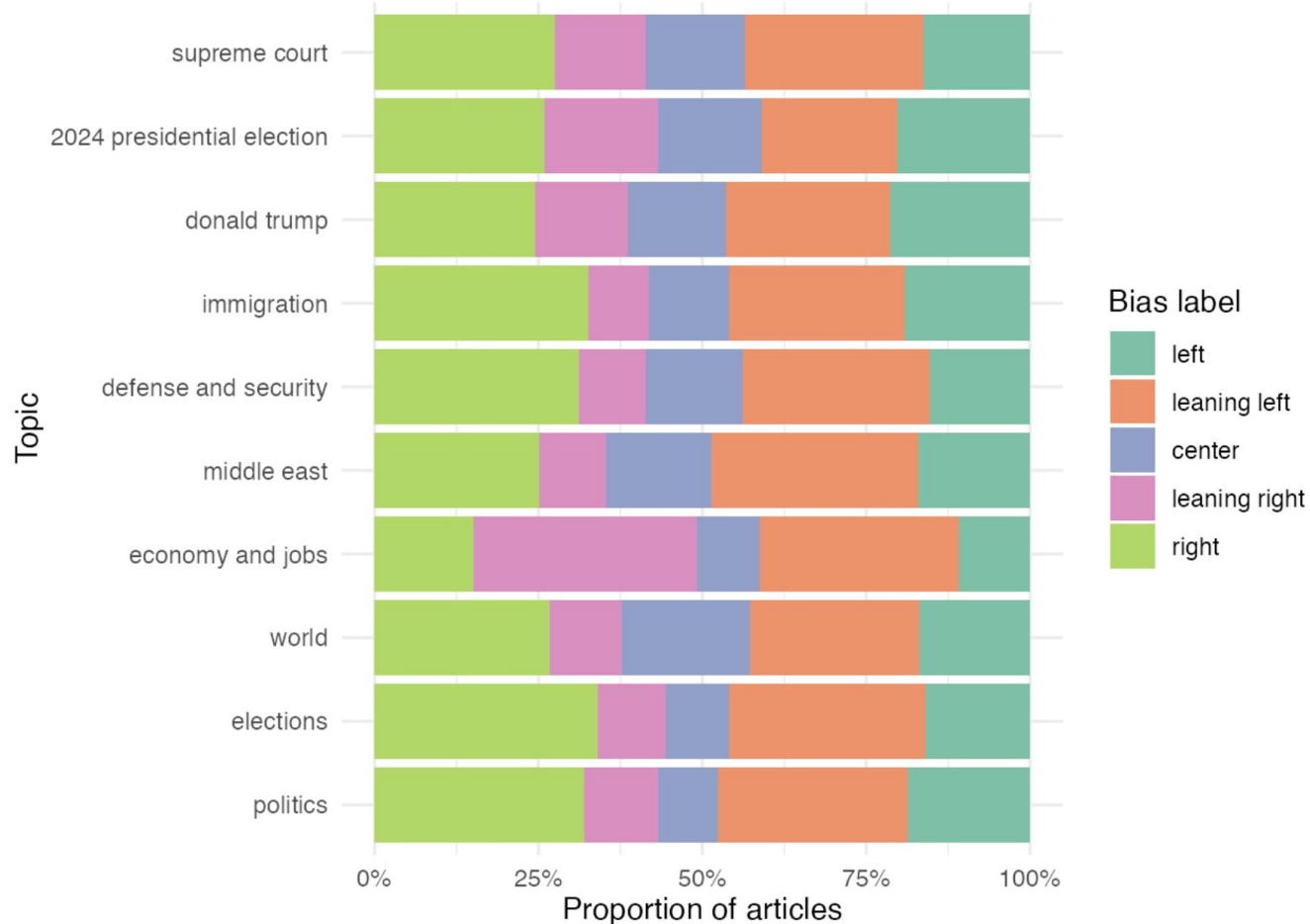
Article Count by Bias Label



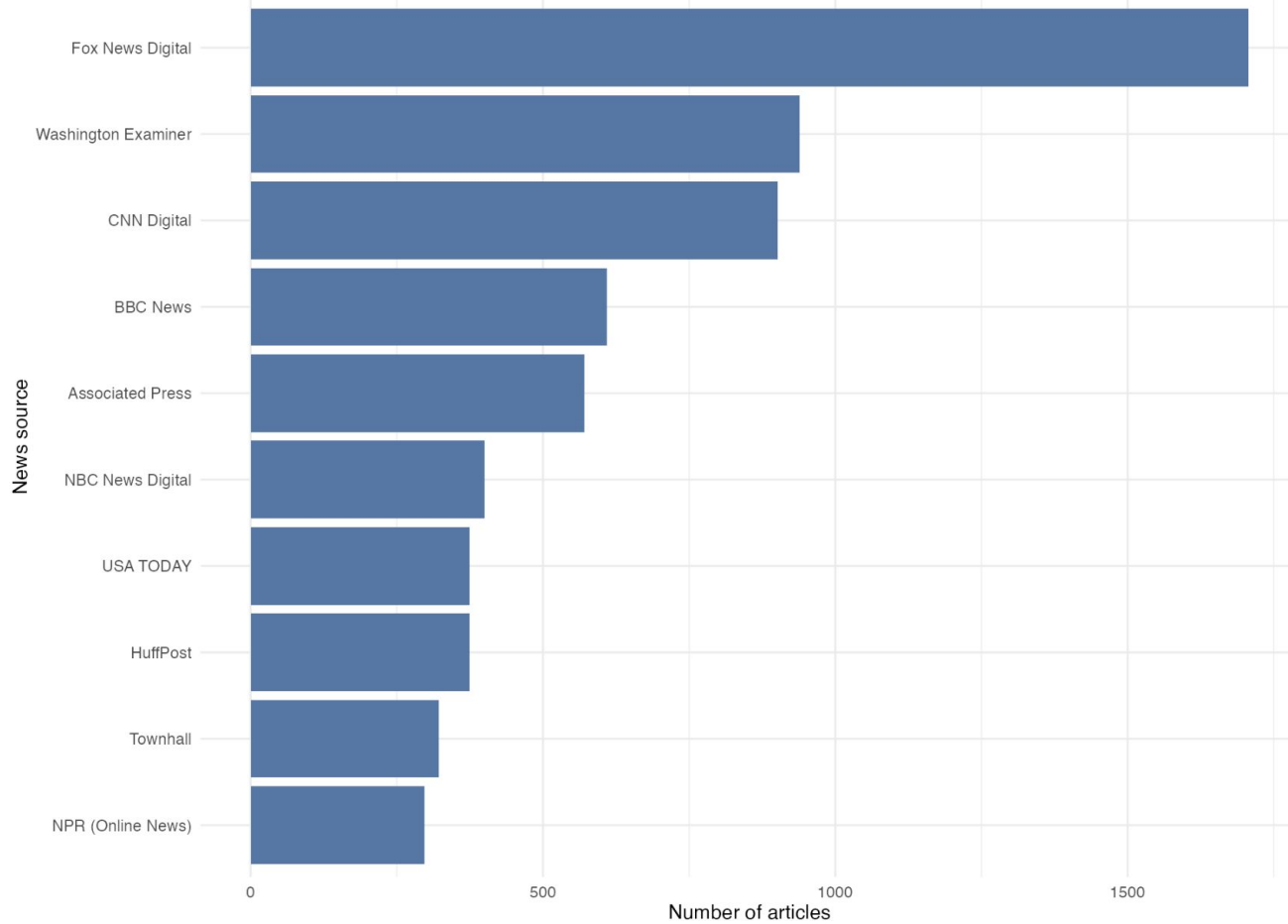
Most Common Topics

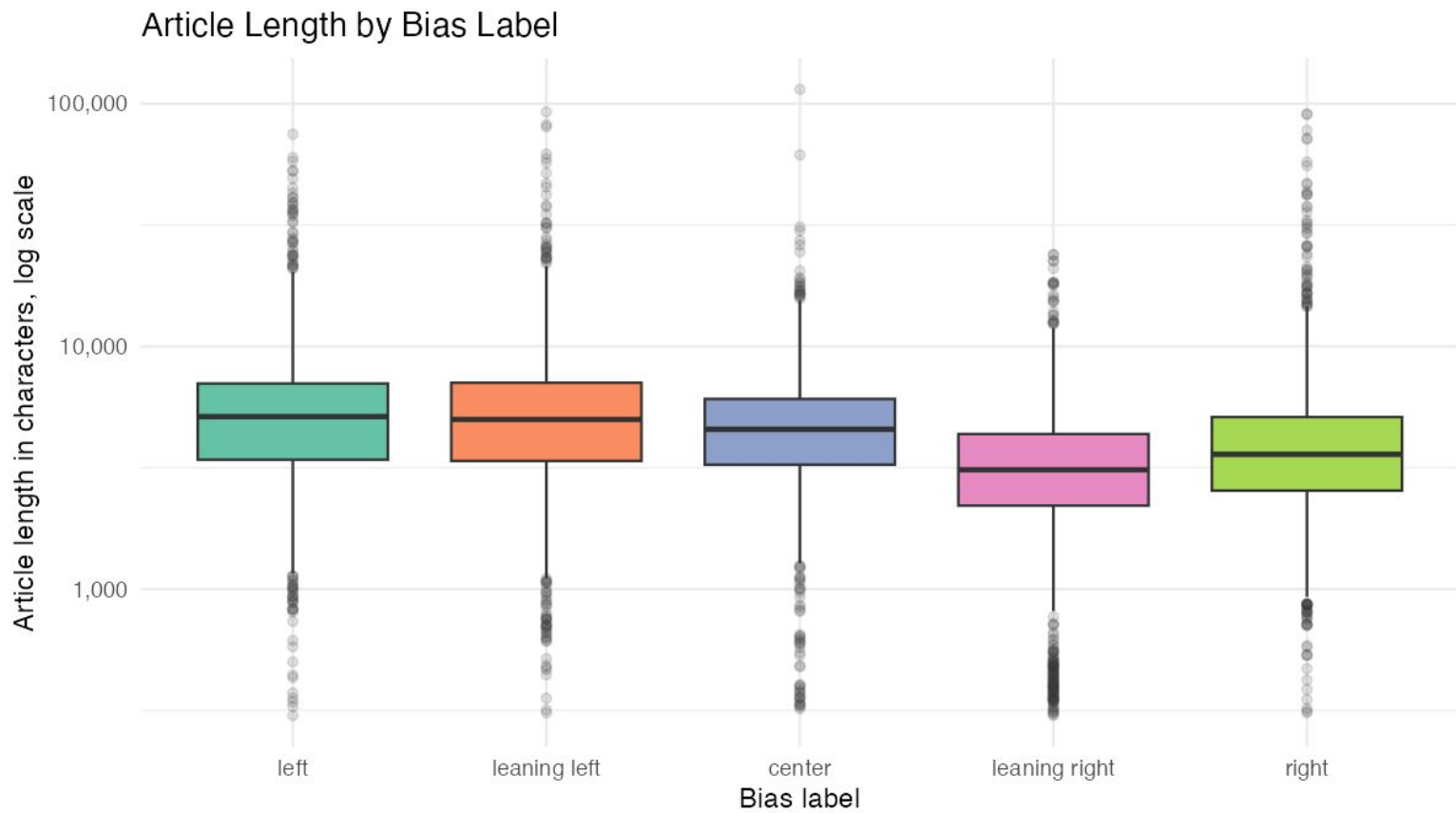


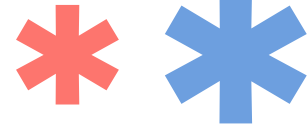
Bias Proportions for Top 10 Topics



Top News Sources by Article Count

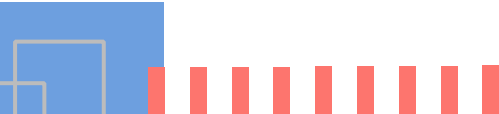






3

Models

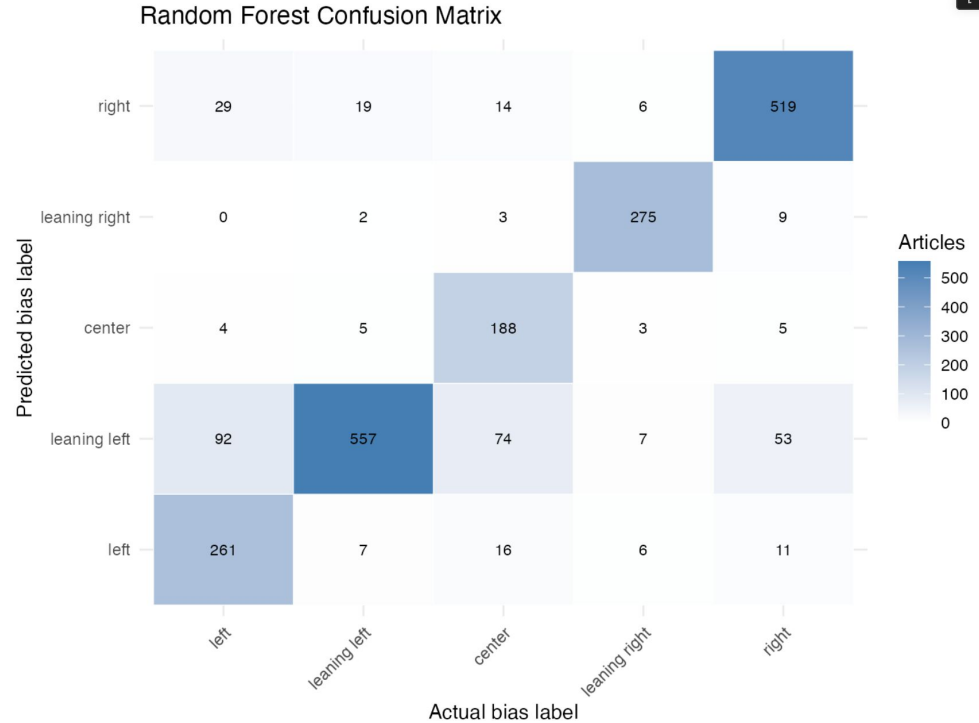


Model Metrics

- **Accuracy**

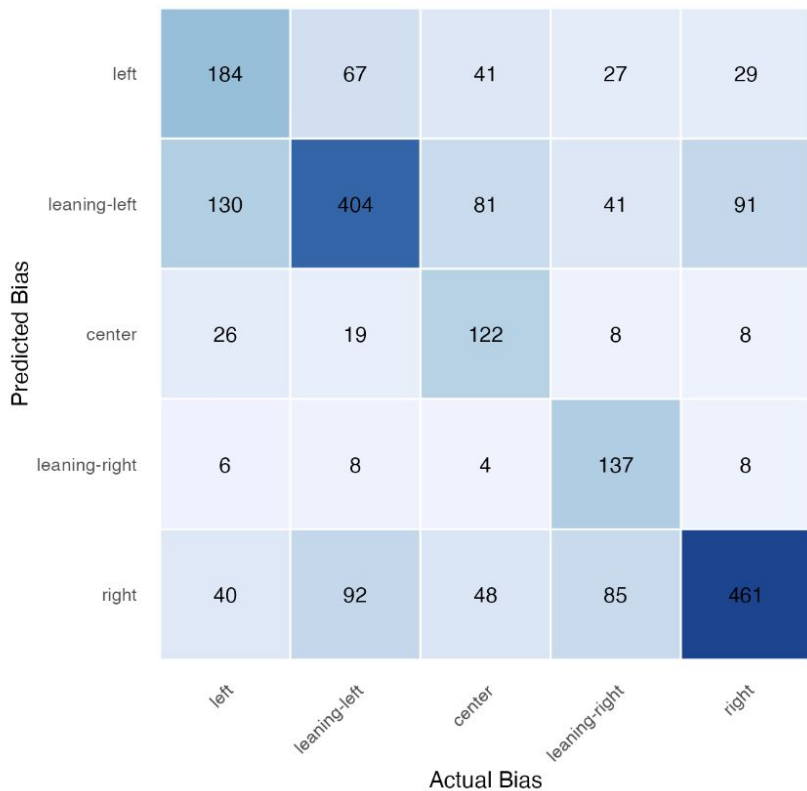
left = 0, leaning left = 1...

- **Mean Squared Error**
- **Mean Absolute Error**
- **F1 Score**



Bag of Words + LASSO

Confusion Matrix

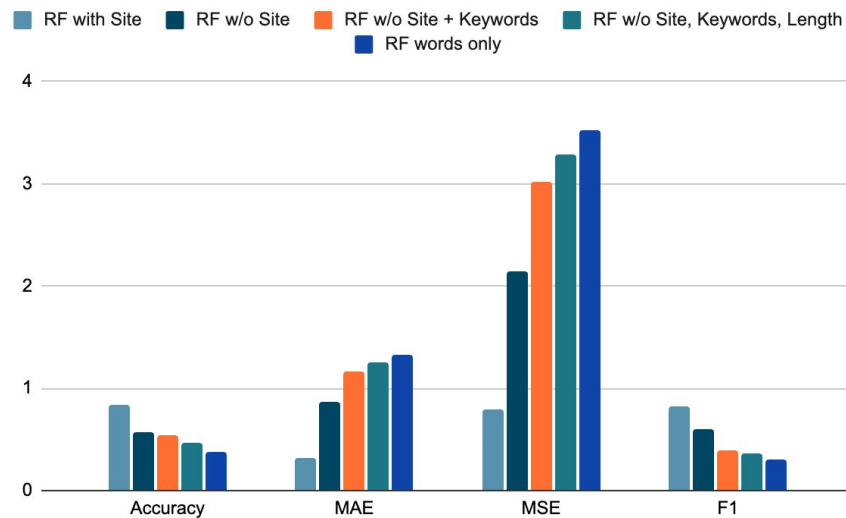
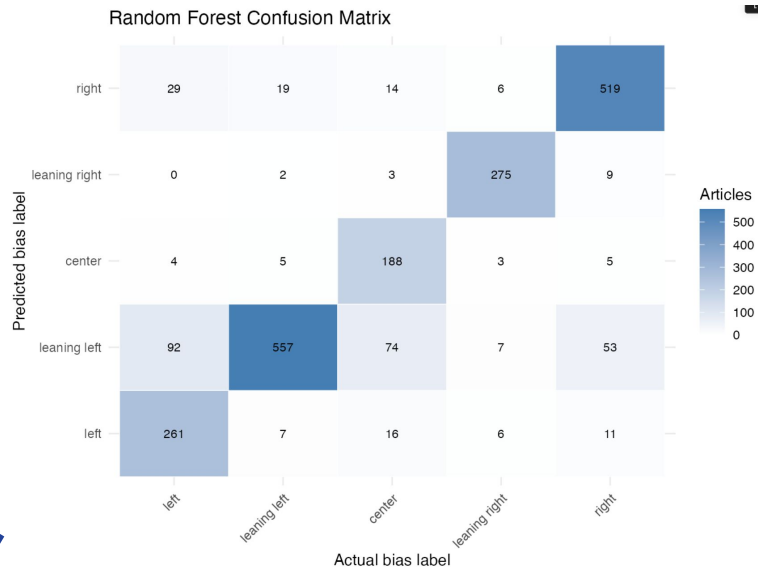


Accuracy : 0.6036
95% CI : (0.5826, 0.6243)
No Information Rate : 0.2755
P-Value [Acc > NIR] : < 2.2e-16

macro_f1	:	weighted_f1	:
dbl	:	dbl	:
	0.5813		0.5972

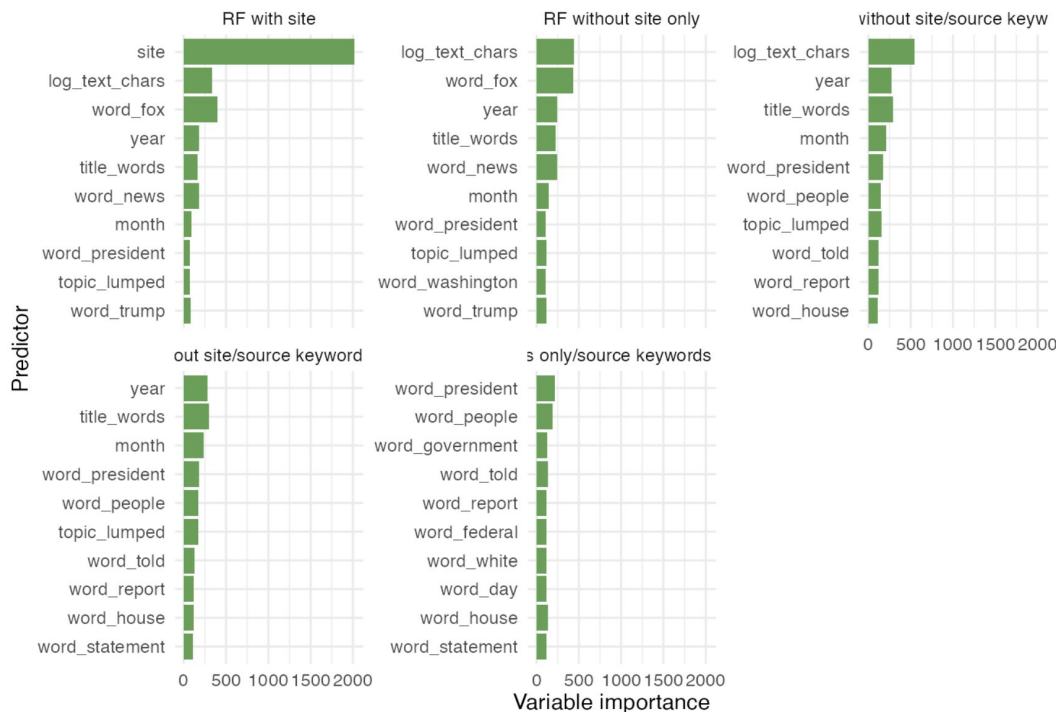
mse	:	mae	:
dbl	:	dbl	:
	1.91		0.7707

Random Forest



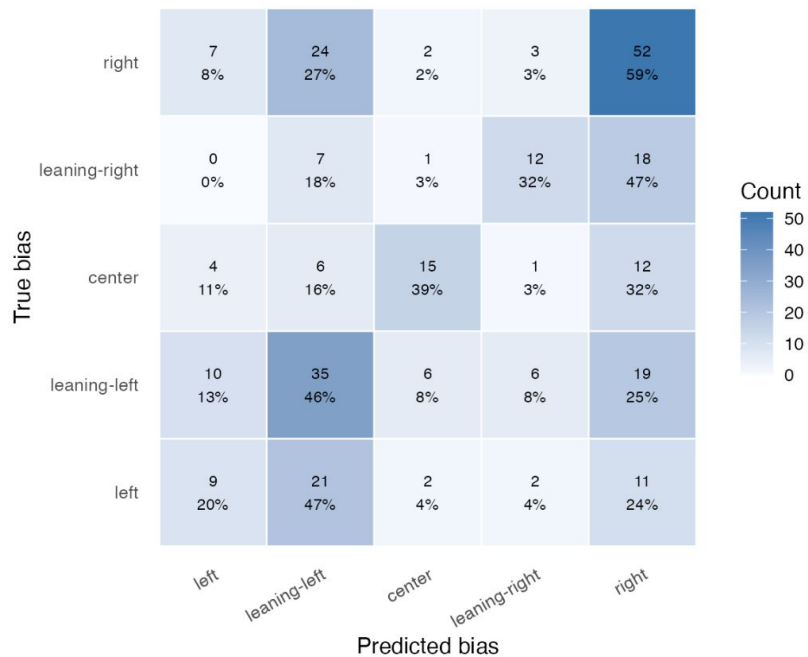
Random Forest

Top Predictors Across Random Forest Models



BERT Embeddings + LASSO

BERT Embeddings + LASSO Confusion Matrix



Accuracy : 0.4316

95% CI : (0.3733, 0.4913)

No Information Rate : 0.3088

P-Value [Acc > NIR] : 8.377e-06

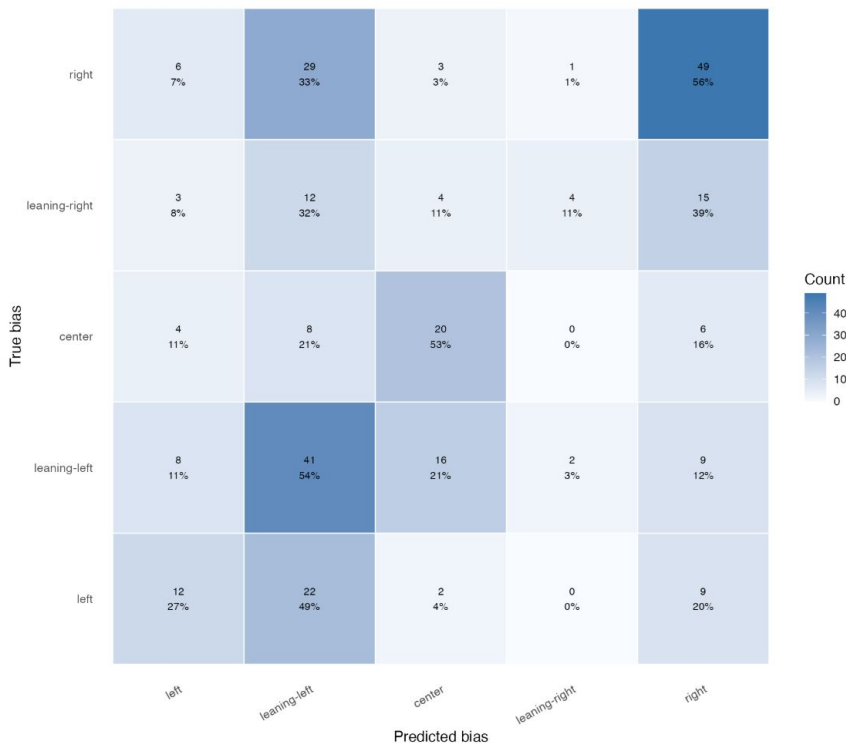
mse	:	mae	:
dbl	:	dbl	:
	3.13		1.19

macro_f1	:	weighted_f1	:
dbl	:	dbl	:
	0.4060		0.4230



BERT Embeddings + Neural Network

BERT Embeddings + Neural Network Confusion Matrix



Accuracy : 0.4421

95% CI : (0.3836, 0.5019)

No Information Rate : 0.3088

P-Value [Acc > NIR] : 1.514e-06

macro_f1 dbl	:	weighted_f1 dbl	:
0.3921		0.4248	

mse dbl	:	mae dbl	:
2.80		1.11	

Fine-tuned BERT (1k)

- Fine-tuned BERT results on the 200-row held-out test set:

Accuracy: 0.57 (95% CI: 0.498–0.640) — well above the 0.20 random baseline for 5 classes, and above the 0.354 majority-class accuracy you'd get from picking "center" every time in a real skewed dataset.

Per-class Precision / Recall / F1:

Class	Precision	Recall	F1
left	0.427	0.725	0.537
leaning-left	0.500	0.325	0.394
center	0.641	0.625	0.633
leaning-right	0.742	0.575	0.648
right	0.667	0.600	0.632

Macro F1: 0.569 (= weighted F1 here since classes are balanced 40/class)

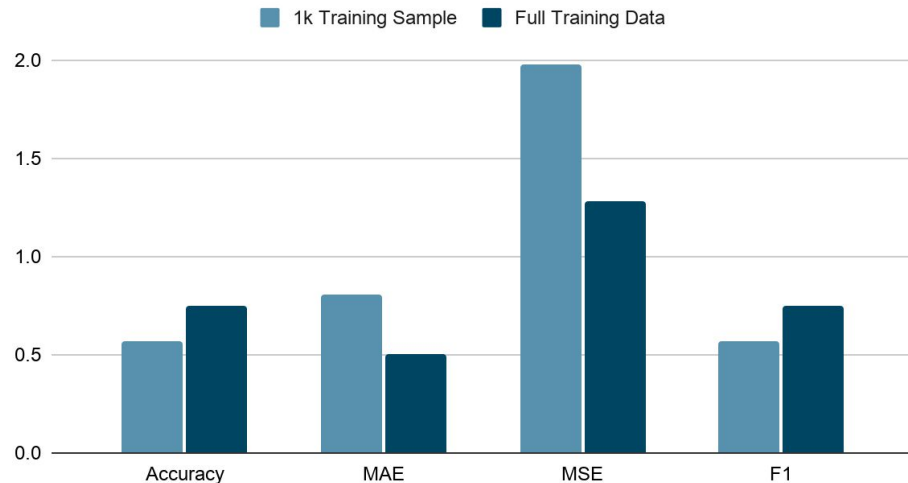
Ordinal error (treating bias as a 0–4 scale, left→right):

- MSE: 1.975
- RMSE: 1.405
- MAE: 0.805

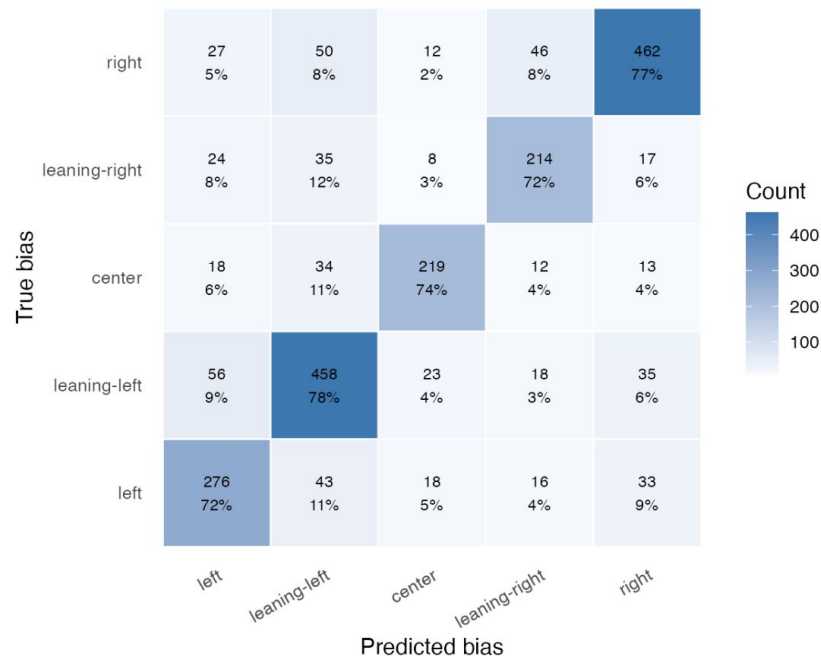
Fine-tuned BERT: 1k vs 8k

Using: google-bert/bert-base-uncased

Fine-tuned BERT Trained on 1k Sample vs Full Training Data

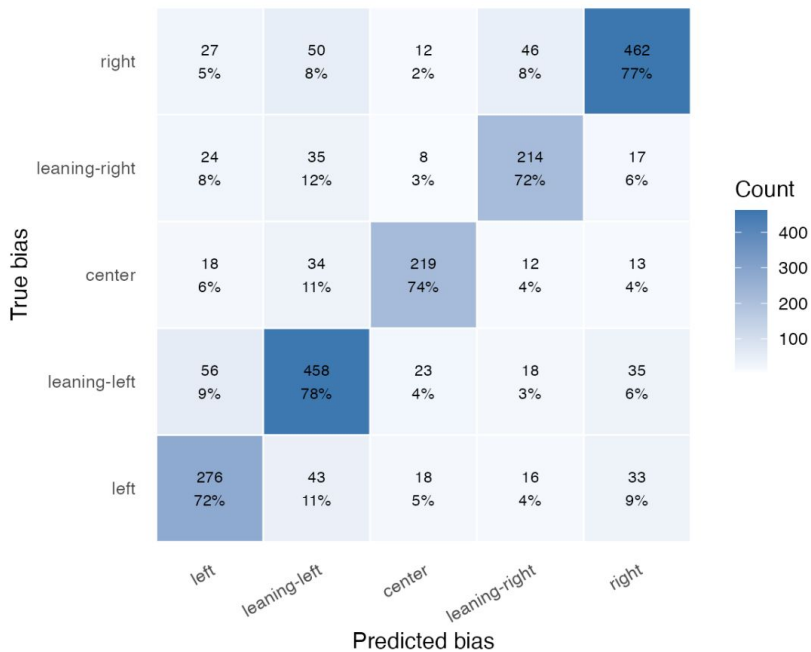


Fine-Tuned BERT Confusion Matrix



Fine-tuned BERT (Whole)

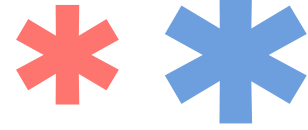
Fine-Tuned BERT Confusion Matrix



macro_f1	:	weighted_f1	:
dbl		dbl	
	0.7452		0.7524

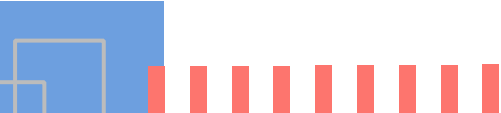
mse	:	mae	:
dbl		dbl	
	1.28		0.4993

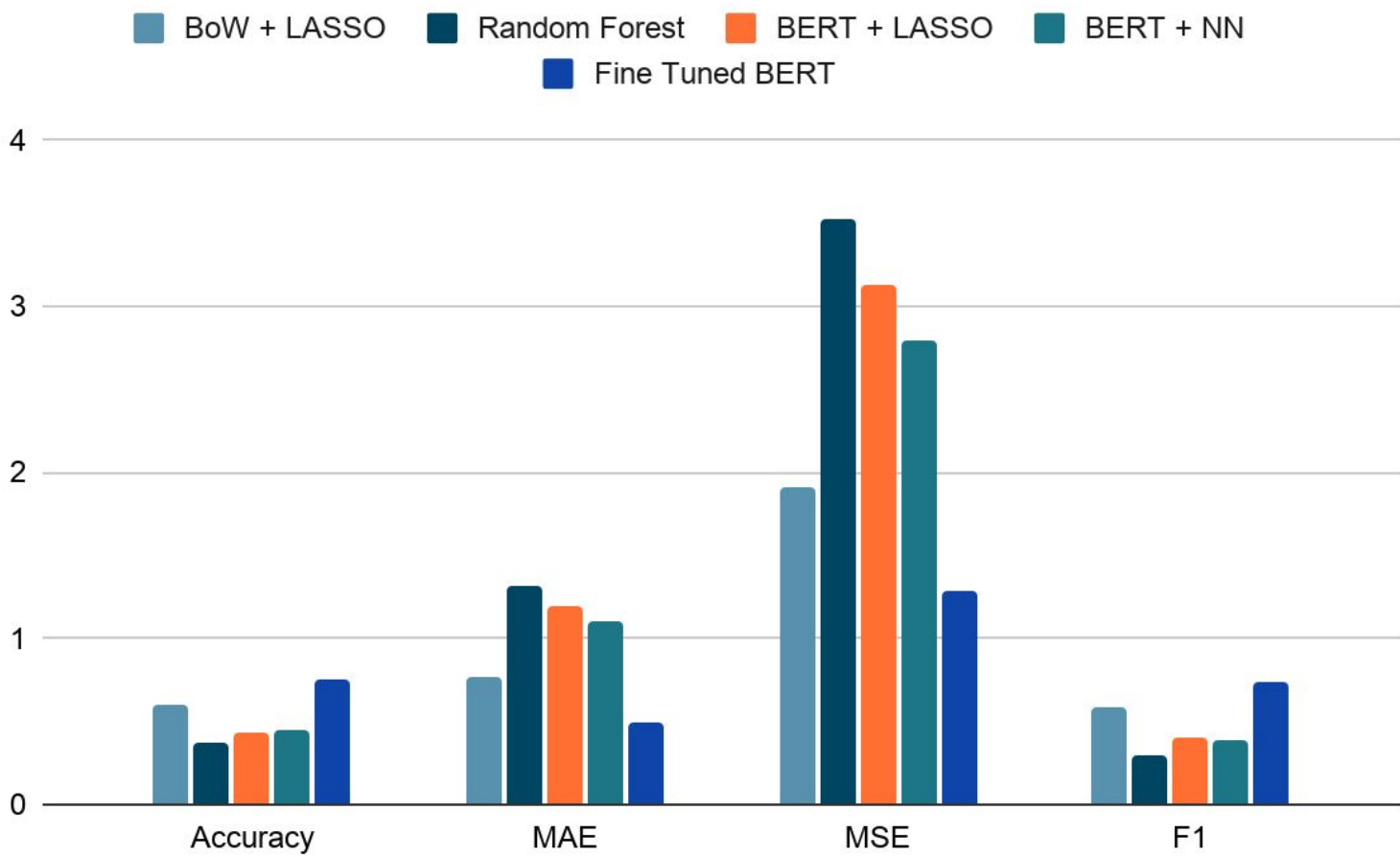
Accuracy : 0.7517
95% CI : (0.733, 0.7698)
No Information Rate : 0.2755
P-Value [Acc > NIR] : < 2.2e-16

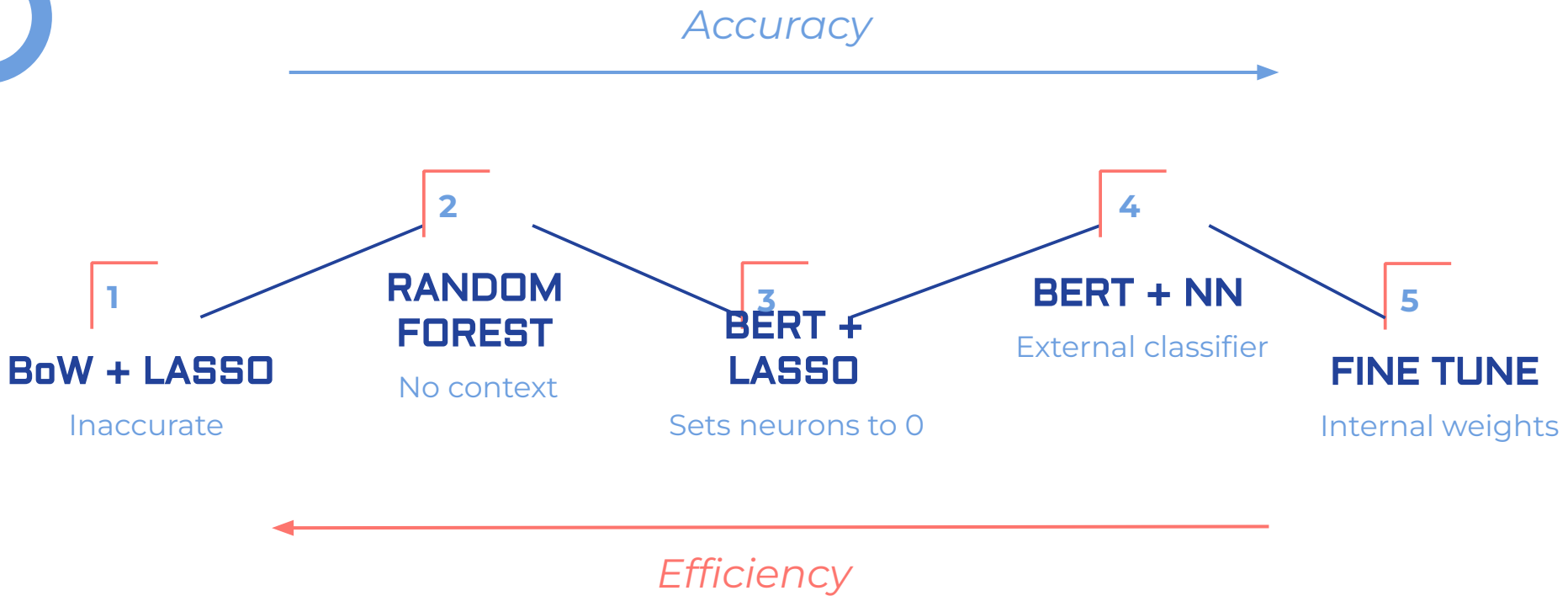


4

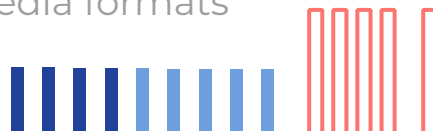
Comparison







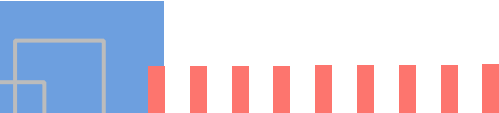
Limitations - selective coverage of events, does not work on other media formats





5

Conclusion





AlterNet

The Atlantic

DAILY BEAST

DEMOCRACY
NOW!

The
Guardian

HUFFPOST

The
Intercept

JACOBIN

Mother Jones

MS NOW

THE
Nation.

The New York Times
(opinion)

THE
NEW YORKER

SLATE

Vox

abc NEWS

AP

AXIOS

Bloomberg

CBS NEWS

CNBC

CNN

INSIDER

NBC NEWS

The New York Times
(news)

npr

POLITICO

PROPUBLICA

SEMAFOR

TIME

USA
TODAY

The Washington Post

yahoo!
news

1440

BBC
NEWS

THE CHRISTIAN SCIENCE
MONITOR.

Forbes

THE
HILL

MarketWatch

MORNING BREW

NEWSNATION

Newsweek

reason

REUTERS

SAN
STRAIGHT-ARROW NEWS

TANGLE

THE WALL STREET JOURNAL.
(news)

The American
Conservative

Daily Mail

THE DISPATCH

THE EPOCH TIMES

FOX
BUSINESS

THE
FREE
PRESS

Just the News.

NATIONAL
REVIEW
(news)

RealClear Politics

UPWARD

THE WALL STREET JOURNAL.
(opinion)

WASHINGTON
Examiner

The
Washington
Times

ZeroHedge

THE AMERICAN
SPECTATOR

Blaze media

BREITBART

CBN

DAILY CALLER

DAILY WIRE

FOX
NEWS

the
FEDERALIST

IJR. INDEPENDENT
JOURNAL REVIEW

NATIONAL
REVIEW
(opinion)

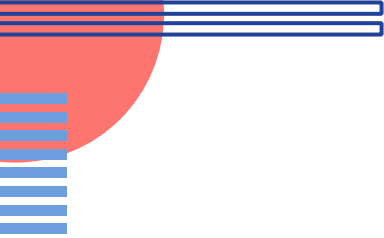
NEW YORK POST

NEWSMAX

OAN
One America News Network

The Post Millennial.

THE WASHINGTON FREE BEACON



AI USAGE

- Help with R syntax
 - Code to clean data
- Help with debugging





THANK YOU!

